1. Probability Theory

1. Basic Definitions

Probability space: $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is sample space, $\mathcal{F}$ is σ-algebra, $\mathbb{P} : \mathcal{F} \to [0,1]$ with $\mathbb{P}(\Omega) = 1$.

Random variable: Measurable $X : \Omega \to \mathbb{R}$.

Expectation: $\mathbb{E}[X] = \int X \, d\mathbb{P}$

Variance: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$

2. Concentration Inequalities

Markov: For $X \geq 0, t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$$

Chebyshev: For $k > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq k) \leq \frac{\mathrm{Var}(X)}{k^2}$$

Hoeffding: For i.i.d. $X_i \in [a,b]$, $\varepsilon > 0$,

$$\mathbb{P}\left(|\overline{X}_n - \mathbb{E}[X]| \geq \varepsilon\right) \leq 2\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right)$$

where $\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$.

Decay rates: Markov $O(1/t)$, Chebyshev $O(1/k^2)$, Hoeffding $O(\exp(-n\varepsilon^2))$.

3. Laws & Limits

Weak Law of Large Numbers:

$$\mathbb{P}\left(|\overline{X}_n - \mu| \geq \varepsilon\right) \to 0 \text{ as } n \to \infty$$

Central Limit Theorem:

$$\frac{\sqrt{n}\left(\overline{X}_n - \mu\right)}{\sigma} \longrightarrow \mathcal{N}(0,1)$$

Key variance fact: $\mathrm{Var}\left(\overline{X}_n\right) = \frac{\sigma^2}{n}$

2. Linear Algebra

1. Norms

Norm axioms: $\|x\| \geq 0$, $\|\alpha x\| = |\alpha| \, \|x\|$, $\|x + y\| \leq \|x\| + \|y\|$.

ℓ_p **norms:** $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}$

- ℓ_1: $\|x\|_1 = \sum_i |x_i|$ (Manhattan)
- ℓ_2: $\|x\|_2 = \sqrt{\sum_i x_i^2}$ (Euclidean)
- ℓ_∞: $\|x\|_\infty = \max_i |x_i|$

Norm equivalence (finite dim): All norms equivalent in $\mathbb{R}^n$.

2. Inner Products

Inner product axioms: Linearity, symmetry $\langle x, y\rangle = \langle y, x\rangle$, positive-definiteness $\langle x, x\rangle \geq 0$.

Standard dot product: $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$

Induced norm: $\|x\| = \sqrt{\langle x, x\rangle}$

Cauchy-Schwarz:

$$|\langle x, y\rangle| \leq \|x\|\|y\|$$

Equality iff $x = \lambda y$ for some λ.

3. Matrices

SVD: $A = U\Sigma V^\top$ where U, V orthogonal, Σ diagonal with singular values $\sigma_1 \geq \ldots \geq \sigma_r > 0$.

Rank: $\mathrm{rank}(A) = r$ = number of nonzero singular values.

Frobenius norm: $\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\sum_i \sigma_i^2}$

Spectral norm: $\|A\|_2 = \sigma_1$ (largest singular value).

Low-rank approx: Best rank-k approx is $A_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$ with error $\|A - A_k\|_2 = \sigma_{k+1}$.

3. Functional Analysis

1. Metric Spaces

Metric: $d : X \times X \to \mathbb{R}_+$ satisfying:

- $d(x,y) = 0$ iff $x = y$
- $d(x,y) = d(y,x)$ (symmetry)
- $d(x,z) \leq d(x,y) + d(y,z)$ (triangle ineq)

Complete: Every Cauchy sequence converges.

Examples: $(\mathbb{R}^n, \|\cdot\|_2)$, $(\mathcal{C}[0,1], \|\cdot\|_\infty)$.

2. Function Spaces

L^p **spaces:** Functions with $\int |f|^p < \infty$.

L^p **norm:** $\|f\|_p = \left(\int |f(x)|^p \, dx\right)^{1/p}$

L^∞ **norm:** $\|f\|_\infty = \text{ess sup } |f(x)|$

L^2 **inner product:** $\langle f, g\rangle = \int f(x)g(x) \, dx$

Completeness: L^p spaces are complete (Banach spaces). L^2 is a Hilbert space.

4. Optimization

1. Convexity

Convex set: $\lambda x + (1-\lambda)y \in C$ for all $x, y \in C$, $\lambda \in [0,1]$.

Convex function: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$.

Strictly convex: Strict inequality for $\lambda \in (0,1)$, $x \neq y$.

Jensen's Inequality:

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

for convex f.

2. Gradients & Optimality

Gradient: $\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)^\top$

First-order optimality (unconstrained):

$$\nabla f(x^*) = 0$$

Second-order sufficient condition:

$$\nabla^2 f(x^*) \text{ is positive definite}$$

Convex first-order: For convex f,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle$$

3. Constrained Optimization

KKT conditions: For $\min f(x)$ s.t. $g_i(x) \leq 0$, $h_j(x) = 0$:

- **Stationarity:** $\nabla f(x^*) + \sum_i \mu_i \nabla g_i(x^*) + \sum_j \lambda_j \nabla h_j(x^*) = 0$
- **Primal feasibility:** $g_i(x^*) \leq 0$, $h_j(x^*) = 0$
- **Dual feasibility:** $\mu_i \geq 0$
- **Complementarity:** $\mu_i g_i(x^*) = 0$

Lagrangian:

$$\mathcal{L}(x, \mu, \lambda) = f(x) + \sum_i \mu_i g_i(x) + \sum_j \lambda_j h_j(x)$$

5. Complexity Theory

1. Asymptotic Notation

Big-O: $f(n) = O(g(n))$ if $\exists c, n_0$ s.t. $f(n) \leq cg(n)$ for $n \geq n_0$.

Big-Omega: $f(n) = \Omega(g(n))$ if $g(n) = O(f(n))$.

Big-Theta: $f(n) = \Theta(g(n))$ if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.

Little-o: $f(n) = o(g(n))$ if $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$.

Little-omega: $f(n) = \omega(g(n))$ if $g(n) = o(f(n))$.

2. Complexity Classes

P: Problems solvable in polynomial time.

NP: Problems verifiable in polynomial time.

NP-complete: Hardest problems in NP; if any is in P, then P = NP.

NP-hard: At least as hard as NP-complete problems.

co-NP: Complements of NP problems.

Common reductions: 3-SAT, vertex cover, clique, knapsack.

6. Key Pitfalls

Hoeffding vs CLT: Hoeffding gives **finite-sample** bounds; CLT is asymptotic. Use Hoeffding for PAC learning.

Norm confusion: ℓ_1 encourages sparsity, ℓ_2 is rotation-invariant. Different norms → different regularization behavior.

Convexity: Jensen applies to convex f; for concave f, inequality flips.

Independence: Hoeffding requires i.i.d. samples; without independence, use McDiarmid or martingale inequalities.

KKT: Complementarity $\mu_i g_i(x^*) = 0$ means either constraint is inactive ($g_i(x^*) < 0$) or multiplier is zero.